

Week 2: Dataset selection, review and planning

Project: Uncertainty-aware drug–protein interaction network with evidence-guided drug repurposing · Repo: github.com/Itsabhirajurs/Trust_DTI

DONE
Scope fixed with the team; data sources selected; ChEMBL schema and key attributes understood; two data rules set; plan locked.

KEY DECISION
Universal scope: all human single-protein targets, reported per protein family. ChEMBL 37 is the primary source.

NEXT
Code the data pipeline: verify, extract and profile both databases, then publish a Stage 0 data report.

1 What TrustDTI is, in two sentences

TrustDTI predicts whether an existing drug is likely to bind a human protein, attaches a **calibrated confidence** to each prediction, and looks up (never predicts) whether that protein is linked to a disease. It then uses both signals to decide what literature to search: supporting evidence when the model is confident, **counter-evidence** when it is not.

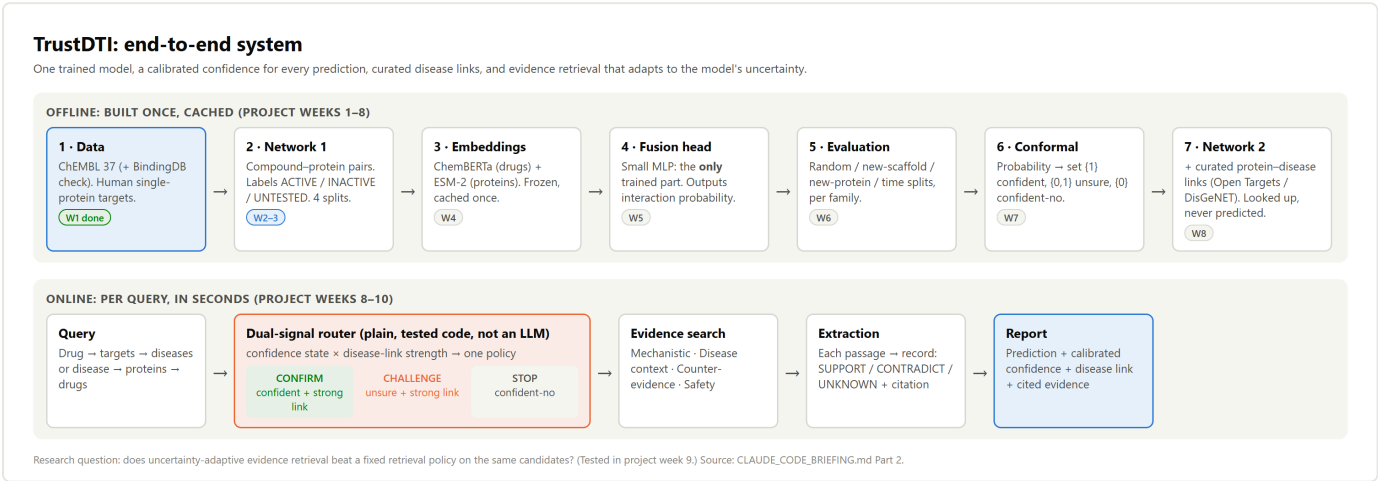


Figure 1. Whole system. This week covers the "Data" stage and its planning.

2 Dataset selection

We needed openly licensed, curated drug–protein measurements with enough detail to judge quality: units, censoring, assay type and origin. **ChEMBL 37** meets all four and is the primary source. **BindingDB** is kept as a cross-check. UniProt supplies protein identity, and the disease and literature sources come in later weeks.

Dataset selection: what each source is for				
Scope: every human single-protein target (universal, reported per protein family). Decided by the team; see DECISIONS.md D-001.				
SOURCE (VERSION)	WHAT IT CONTAINS	ROLE IN TRUSTDTI	USED IN (PROJECT WEEK)	DECISION / WHY
ChEMBL 37 SQLite, 5.8 GB → 30 GB	24.5M curated bioactivity measurements, 2.9M compounds, 18.5k targets, from papers, patents and screening centres.	Primary source. Drug–protein measurements that build Network 1 and train the model.	W1–W7	Selected Largest curated, openly licensed source. Records units, censoring, assay confidence and origin for every value.
BindingDB 202609 TSV, 0.6 GB → 9 GB	3.2M binding measurements (Ki, IC50, Kd, EC50) from literature, patents, PubChem.	Cross-check and possible extra coverage.	W1 check	Review Profiling showed it largely copies ChEMBL (and vice versa). Recommendation: ChEMBL-only for v1.
UniProt Swiss-Prot 2025_03 via ChEMBL	Canonical protein IDs (accessions) and amino-acid sequences.	Protein identity key across databases; sequences for ESM-2 embeddings.	W3–W4	Selected Standard protein identifier; already linked inside ChEMBL.
Open Targets / DisGeNET	Curated gene/protein–disease associations with evidence scores.	Disease anchor (Network 2): looked up, never predicted.	W8	Planned Curated and auditable. Keeps the disease link independent of the model.
PubMed / PubTator NCBI E-utilities	Biomedical abstracts with pre-tagged drugs, genes and diseases.	Literature evidence for the retrieval agent.	W9	Planned Free API with entity annotations, so no custom NER is needed.

Identifiers used for joining: compounds by InChIKey (not SMILES, whose text differs between databases); proteins by UniProt accession.

Figure 2. Each source, its role in the project and the reason for choosing it.

All human single-protein targets, reported per protein family, rather than kinases only. **Gained:** a general, repurposing-relevant network. **Given up:** the kinome's density and cleaner cross-protein comparisons. **Mitigation:** every result is stratified by protein family, and a minimum-data rule per protein is set next.

3 Understanding the data

A single ChEMBL measurement only becomes useful once it is joined to its assay, target, protein and family. We mapped the 11 tables involved and the keys that connect them.



Figure 3. Join path from one measurement to its drug, protein and protein family.

ATTRIBUTE	MEANING	WHY IT MATTERS
standard_type	What was measured (IC50, Ki, Kd, EC50, Potency...)	Only some endpoints are comparable potencies
standard_relation	Operator on the value (=, >, <)	">" means a lower bound, not an exact value
standard_value / units	Normalised value, e.g. 30000 nM	Meaningful only with relation and units
pchembl_value	-log10 molar potency (7 = 100 nM)	Comparable scale; empty for censored values
activity_comment	Free-text outcome ("Not Active", "inconclusive")	Main source of measured-inactive labels
confidence_score	How exactly the target is assigned (0–9)	Target certainty filter
src_id	Who deposited the record	Source affects potency and overlaps
accession / InChIKey	UniProt protein ID / compound key	Joining across databases (not SMILES)
docs.year	Publication year	Needed for the time-based split

Full dictionary (21 fields, with examples): W2_06_attribute_dictionary.csv .

4 Two rules that protect the whole project

Two data rules that protect the whole project
Both prevent silent errors that would make a drug–protein model look better (or worse) than it really is.

Rule 1: every drug–protein pair has exactly one of three states

ACTIVE
A measurement exists **and** shows meaningful activity (threshold set from real data in week 2 of the project plan).

INACTIVE
A measurement exists **and** shows no meaningful activity under the tested conditions, e.g. "not active", IC50 > 10 µM.

UNTESTED
No measurement, or an unusable one ("inconclusive", "not determined").
Never converted to INACTIVE: most drug–protein pairs were simply never tested.

Rule 2: censored values are not exact values
Real row, ChEMBL activity 884645: imatinib vs FLT3, IC50 > 30,000 nM.
It means "no activity found up to 30 µM" (a lower bound), **not** "potency = 30 µM".
Imatinib activity rows in ChEMBL:

All rows: 4,878 (100%)

833 kept by the pChEMBL filter (17%)

A common shortcut (keep only rows with a pChEMBL value) keeps 17% and drops **all 1,335 censored rows**, which are exactly the "inactive" evidence we need.

What this means for the pipeline

- Keep every relation: = > < >= <= >> <<
- Never filter on pchembl_value IS NOT NULL
- "inconclusive" (1.73M rows) is UNTESTED, not INACTIVE
- Enforced by automated tests (tests/test_chembl_extract_data.py)
- Stated openly as a positive-unlabelled problem if negatives turn out scarce

Sources: CLAUDE_CODE_BRIEFING.md §3.3–3.4; notebook 01_chembl_first_look.ipynb; docs/chembl_schema_notes.md.

Figure 4. Three-state labels, and why censored values must be kept. Numbers are from real ChEMBL rows for imatinib.

5 Plan

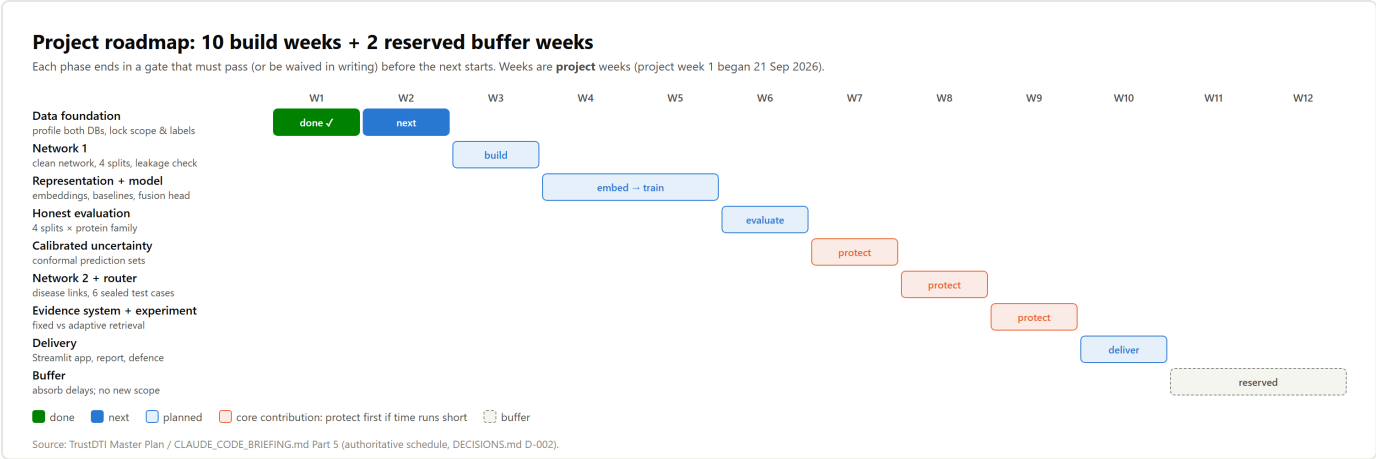


Figure 5. Roadmap in project weeks. Calibration, the router and the central experiment are protected first if time runs short.

DECISION (DATED IN DECISIONS.MD)	WHAT WE CHOSE
D-001 Scope	All human single-protein targets, reported per family
D-002 Schedule	10 build weeks + 2 reserved buffer weeks, with a gate at the end of each phase
D-003 Compute	No local GPU: embeddings on Colab/Kaggle/HPC; free open-weight LLM for the evidence stage
D-005 Data versions	ChEMBL 37 and BindingDB 202609, each checksum-verified

Challenges this week

The ChEMBL bulk download (5.8 GB) failed repeatedly and left a corrupt partial file. We fixed it by resuming on another network and accepting the file only after its checksum matched. Our planning documents also disagreed on scope and schedule; the team settled both and recorded the outcome in DECISIONS.md.

Next week

Verify and extract both databases; count proteins, compounds and pairs per family; check sources, quality and missing values; measure ChEMBL–BindingDB overlap; publish the Stage 0 data report with an open-issue list.

Supporting files in this folder: W2_01–W2_05 (figures), W2_06_attribute_dictionary.csv, references.md, WEEK02_FORM_ENTRIES.md. Figures are built from source by docs/weekly_reports/_build/.